

MOHAMMED GHALIB

AI Engineer • RAG & Agentic Systems • LLM Applications in Production

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[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

AI engineer who builds LLM systems that are reliable enough to ship. I design retrieval and agentic pipelines that plan their own lookups, ground every answer in real sources, verify their own output, and refuse to guess when they are not sure. My focus is the hard part of applied AI: making a model trustworthy in front of real users, and proving it with evaluation rather than guesswork.

I have shipped this work as the core of two live products. Sheria AI, a Kenyan-law assistant, runs an agentic LangGraph pipeline: triage, a research step that plans its own tool lookups over case-law and constitutional corpora, and a draft, verify, and repair loop, backed by triage and blind side-by-side evaluation harnesses. Ummly, an Islamic knowledge platform, runs a Pinecone RAG assistant with reranking, a confidence floor, query rewriting, and bilingual answers with per-excerpt citations.

I build the whole system around the model, not just the prompt. I write the Django and FastAPI backends, the React and React Native apps, and I containerise, deploy, and maintain them in production. That means I can take an AI feature from idea to a live, monitored service on my own.

CORE SKILLS

AI & LLM Engineering: Agentic pipelines on LangGraph (plan, research, verify, repair), tool-calling / function-calling, RAG design, prompt engineering, intent routing, hallucination control, grounding & source citation, graceful refusal on low confidence, evaluation harnesses (triage scorecards, blind side-by-side A/B)

LLMs & Providers: OpenAI, LLaMA (via Groq), Google Gemini, DeepSeek, reasoning models (gpt-oss), multi-model routing, tiering, and fallback orchestration

Retrieval & Vectors: Pinecone, ChromaDB, Sentence-Transformers embeddings, chunked indexing, reranking (bge-reranker), similarity-threshold retrieval, hybrid lookup (vector search + exact match), dual-corpus retrieval

AI Frameworks & Serving: LangGraph, LangChain, Sentence-Transformers, token streaming over server-sent events (SSE), Celery for async tasks, prompt-as-config (behaviour tuned in editable prompt files)

Backend & APIs: Python, Django, Django REST Framework, FastAPI, REST, WebSockets/Channels, JWT, PostgreSQL, Redis, SQLite

Frontend: React, React Router (SSR), TypeScript, React Native (Expo), Tailwind CSS, Redux Toolkit, TanStack Query, streaming chat UIs

Infra & Delivery: Docker & docker-compose, Gunicorn, Nginx (nginx-proxy + Let's Encrypt), DigitalOcean, Vercel, GitHub Actions CI/CD, SonarQube, PyTest & Vitest, versioned releases

PROFESSIONAL EXPERIENCE

Software Engineer, Datawise Africa

2026 • Remote

Kenyan AI company. I own an AI legal product end to end and build AI systems for clients and internal teams.

- Own **Sheria AI** ([sheria.africa](#)), an AI legal assistant for Kenyan law, from architecture to live deployment. Rebuilt its core from a single retrieve-then-answer call into an **agentic pipeline on LangGraph**: each question is triaged, then researched by a model that plans its own lookups against case-law and Constitution tools, and each answer is drafted, verified, and repaired before the user sees it, with a coverage check so no part of a multi-part question is dropped.
- **Engineered the retrieval layer.** Chunked the case index so search reads the court's actual reasoning instead of just the header, tuned the relevance thresholds that keep irrelevant cases out of answers, added exact-name case lookup, and enforced grounded citations with a firm no-fabrication rule and graceful refusal when nothing relevant is found.
- **Route work across model tiers:** a reasoning model for triage and tool-planning, a separate model chosen by blind A/B for the user-facing writing, and a lightweight fallback that keeps a turn alive when a tier fails.
- **Built the evaluation harnesses** the team ships on: a triage scorecard, and a blind side-by-side comparison of candidate answer models. Prompt and model changes are backed by evidence, not guesswork. The judgment lives in editable prompt files, so behaviour is tuned in prose rather than code.
- **Built the streaming chat UI** (React Router SSR) that renders answers token by token over server-sent events, with ranked source cards, and deliver client and internal AI systems beyond Sheria AI.

Independent AI & Software Engineer (Contract), Private Clients

Jan 2024 – Aug 2025 • Remote

AI products and full-stack platforms designed, built, and deployed for clients.

- **Ummly (formerly Islamify), Islamic knowledge platform** (ummly.co.ke). Built solo a **Pinecone RAG** assistant grounded in the Qur'an and Hadith: server-side embeddings, **bge-reranker reranking**, a confidence floor that returns a low-confidence message instead of guessing, intent classification, query rewriting, bilingual EN/AR answers, and per-excerpt citations. LLaMA via Groq with a Google Gemini fallback. Shipped the full product around it: Django / DRF backend, a React Router (SSR) web app, and an Expo / React Native mobile app, containerised and deployed via GitHub Actions to a DigitalOcean server.
- **Also delivered non-AI full-stack platforms** for clients: Collaborative Approach, a referral and tour-booking SaaS with M-Pesa, PayPal, and Stripe payments, and a multi-portal school management system, both on React, Django, and PostgreSQL with CI/CD.

Software Engineer, Digital Appex

2025 • Remote

Kenyan technology company delivering software and AI. I lead client delivery, including AI solutions.

- **Deliver AI solutions for clients** across healthcare and education, turning requirements into shipped systems, and built the AI chatbot integrated into the company platform.
- Built and now own digitalappex.co.ke **end to end** (Django / DRF backend, React frontend) including its AI chatbot, and keep it live in production with payments, role-based access, background jobs, Docker, and GitHub Actions CI on a DigitalOcean server.

Software Engineer, Bee MultiScent Ltd

Feb 2023 – Dec 2023 • Hybrid

Scent-technology company. I ran the AI diffuser work, the internal platform, and the website.

- **Developed the AI for a smart scent diffuser**, automating how and when the company's IoT diffuser released scent.
- **Built the company portal** (Django / DRF backend, React frontend, real-time dashboards, WebSocket notifications, companion React Native app) and the company website.

Financial Systems Intern, Kenya Commercial Bank (KCB)

Jan 2025 – Jul 2025 • On-site, Nairobi

Production support and integrations for Oracle Fusion at one of Kenya's largest banks.

- Supported Oracle Fusion HCM and Financials for 1,000+ staff, governed roles and security policies (RBAC), and built secure REST integrations with internal services and external systems including KRA iTax and payment gateways.

EDUCATION

Bachelor of Business Information Technology, Mount Kenya University

2022 – 2025

Major in Information Technology, Software Development, and Artificial Intelligence.

CERTIFICATIONS

- Build AI-Powered Apps, Code with Mosh
- Ultimate Django Series, Code with Mosh
- Ultimate Docker Course, Code with Mosh
- Ultimate React + TypeScript Course, Code with Mosh

REFERENCES

Available on request.